

Prakhar Batwal

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Education

Parul University

B.Tech in Computer Science & Engineering

Vadodara, India

Aug 2023 – Apr 2027

Technical Skills

- **Languages:** Java, JavaScript (ES6+), Python, SQL (MySQL, PostgreSQL)
- **Frameworks & Libraries:** Spring Boot, Node.js, Express.js, React.js, Socket.io, Tailwind CSS, Redux
- **Databases & Caching:** MongoDB, Redis, MySQL — transactions, indexing, aggregation pipelines
- **DevOps & Tools:** Git, Docker, Postman, Linux, RESTful APIs, JWT, OAuth 2.0
- **CS Fundamentals:** DSA, OOP, DBMS, OS, Computer Networks, System Design

Projects

High-Throughput Notification Engine — Event-Driven System

Jun 2026 – Present

Java 17 · Spring Boot 3 · Apache Kafka · Redis · MySQL · Docker

GitHub

- Engineered a priority-tiered message processing pipeline using **Apache Kafka** (partitioned topics for urgent transactional alerts vs bulk notifications) to handle traffic bursts without message loss.
- Implemented a distributed rate-limiter and duplicate prevention filter using atomic **Redis Lua scripts** (*SETNX* logic), protecting system integrity on high-value transaction routes.
- Programmed automatic failure recovery utilizing **Exponential Backoff** with multiplier delays, routing failed messages to a Dead Letter Queue (DLQ) for automated diagnostics.
- Modeled a delivery state machine in MySQL, leveraging Redis read-through caching to keep status lookups under **50ms**.

CollabMind — Real-Time Collaborative Project Hub

Nov 2025 – Jan 2026

React.js · Node.js · Express.js · MongoDB · Redis · Socket.io

GitHub | Live Demo

- Delivered **sub-100ms canvas sync latency** across concurrent clients by designing a Socket.io room-scoped broadcasting strategy for real-time collaborative whiteboards and team messaging.
- Reduced analytics dashboard response times by **60%** via MongoDB compound indexes and a Redis read-through cache on active project logs, sustaining 50+ concurrent users without degradation.
- Built a git-inspired differential version-history engine that stores only deltas between saves, cutting document storage footprint by ~70% versus full-snapshot approaches.
- Secured 25+ REST endpoints with JWT + RBAC middleware, clearing full OWASP A01 broken-access-control checks.

AushadhSanchaya — Clinical Ledger & Pharmaceutical Supply Chain

Mar 2026 – May 2026

React.js · Node.js · Express.js · MongoDB · Gemini AI · Tailwind CSS

GitHub | Live Demo

- Guaranteed **100% ledger consistency** across multi-item flows via MongoDB Session transactions with atomic commit/rollback, eliminating dual-write inconsistencies under concurrent API load.
- Cut expired-medicine waste by **35%** by engineering a FEFO batch-tracking engine that auto-routes inventory picks to the nearest expiry date, replacing manual sorting across all procurement cycles.
- Automated incoming stock registration with **Gemini 1.5 Pro Vision OCR** to parse package labels and match items against open purchase orders, removing manual data entry and cutting intake errors to zero.
- Hardened 20+ API endpoints with **hierarchical RBAC** (JWT cookies + Passport.js OAuth), reducing unauthorized-access surface to zero across all security reviews.

Achievements & Profiles

- **Competitive Programming:** Solved 400+ DSA problems on LeetCode & CodeChef; consistently targets medium-hard difficulty with sub-500ms runtime solutions in Java.
- **Elevate Hackathon — Top 10 of 700+ teams:** Shipped a production-ready application in 48 hours, placing in the **top 1.4%** of participants.
- **Engineering Leadership:** Led a 4-person team across 2 projects — defined Git branching strategy, reviewed PRs, and delivered both on schedule with zero post-launch critical bugs reported.
- **YouTube Tech Channel:** Publishes full-stack coding tutorials documenting real project builds, strengthening technical communication and documentation skills alongside engineering practice.

Relevant Coursework

Data Structures & Algorithms · Object-Oriented Programming · Database Management Systems · Operating Systems · Computer Networks · Software Engineering